

Exercise 1

Problem 1: Measuring flat universes

In the following consider two flat ($k = 0$), expanding universes: one that is matter dominated, i.e. $\Omega_m = 1$, and one that is dominated by a cosmological constant, i.e. $\Omega_\Lambda = 1$.

- (a) Derive for both universes an expression for the comoving distance

$$D(z) = \omega(z)R_0$$

as a function of redshift z .

- (b) The "observable universe" is the part of the universe up to redshift $z = \infty$. Compute for both of our universes the comoving volume of this region, i.e. the volume of the corresponding region on the spatial hypersurface of constant time τ_0 . Is there a qualitative difference between the two volumes? What is the meaning of the "observable universe" in the context of causality?

Hint: The spatial volume element for the Robertson Walker (RW) metric at time τ is

$$dV = R^3(\tau)S_k^2(\omega) \sin \theta d\theta d\phi d\omega$$

Problem 2: Change of redshift with time

The redshift of a given source of light should change with time and in principle we could detect this effect by observing the source over a certain timespan. Assume a flat concordance cosmology with $H_0 = 73 \text{ km}/(\text{s} \cdot \text{Mpc})$, $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$. How much would the redshift of a source at redshift $z = 1$ change within ten years? Do you think this might be observable in practice and what complications might arise?